Contribution Results: Indexed Retention, Slabs and Pinned Metadata

CS5290 Project Group

I. Progressive mechanisms

IA/IF add indexed lists and ownership lookup with adaptive-64/fixed-2 retention. SA/SF add 64 KiB MRAM spans for 2049–4096 B requests. PA/PF additionally pin a 528 B upper buddy prefix in WRAM, costing a measured 540 B endpoint increase. Payload heap, input and empty-byte budget are equal within each comparison.

TABLE I
Final work runtime (milliseconds) for every stage on five complementary cases.

Case	B	IA	IF	SA	SF	PA	PF
Larson 8–4095 B (4T)	160.089	164.537	159.826	24.026	19.894	19.172	16.255
CloudPhysics cache (4T)	122.773	122.925	122.784	114.586	114.528	113.023	113.075
Larson 8–8191 B (4T)	320.457	342.572	331.666	216.369	232.613	204.807	188.202
Gowalla churn (4T)	148.847	26.172	24.961	26.918	25.830	26.954	25.827
PrIM selection (4T)	14.134	14.180	14.180	14.135	14.135	14.133	14.133

B: scan fixed-2; I: indexed; S: indexed plus 64 KiB slabs; P: indexed plus slabs plus pinned metadata. A uses adaptive interval 64; F retains at most two empty spans/class. All rows use four tasklets.

TABLE II
Final paper-derived cases: the baseline and both full-stack policies.

Case	B work (ms)	PA work (ms)	PF work (ms)	PA speedup	PF speedup	PF total speedup
Gowalla linked blocks (4T)	19.330	15.479	15.198	1.249	1.272	1.335
Gowalla growing arrays (4T)	5.752	5.845	5.494	0.984	1.047	1.070
KV growth replay (4T)	24.382	19.657	19.625	1.240	1.242	1.210
32 B microbenchmark (1T)	0.875	1.077	0.990	0.813	0.884	0.799
32 B microbenchmark (16T)	6.938	5.719	5.648	1.213	1.228	1.105
256 B microbenchmark (1T)	2.076	2.005	1.925	1.035	1.079	0.994
256 B microbenchmark (16T)	34.201	27.750	27.641	1.232	1.237	1.192
4 KiB microbenchmark (1T)	20.912	1.858	1.778	11.255	11.763	1.512
4 KiB microbenchmark (16T)	508.474	22.335	22.243	22.766	22.860	6.746

Graph cases are reduced ports; KV is an allocation-only replay and excludes attention/serving. Original request sizes and 128 microbenchmark requests/tasklet are retained. Values below one are regressions.

II. Attribution and costs

The 4 KiB frontend is new: original classes stop at 2 KiB. The largest slab gains therefore reflect caching a previously uncached size, not evidence that every original class should use bigger pages. Fixed retention is faster on many stable traces. The one-tasklet 32 B case regresses; sparse survivors can reserve 128 KiB for 8 KiB of live payload. Pinning gains below isolate metadata placement with the same slab/policy.

III. Scope and validation

The final main comparison has 98 configurations over fourteen settings, all successful. The parent and paper extension audits preserve 47 and 42 native cases respectively. Measurements use a deterministic 350 MHz single-DPU simulator; they do not establish hardware or complete-application throughput. Graphs are reduced Gowalla ports, and KV replay excludes attention and serving. Source, traces, native evidence and simulator hashes are audited.

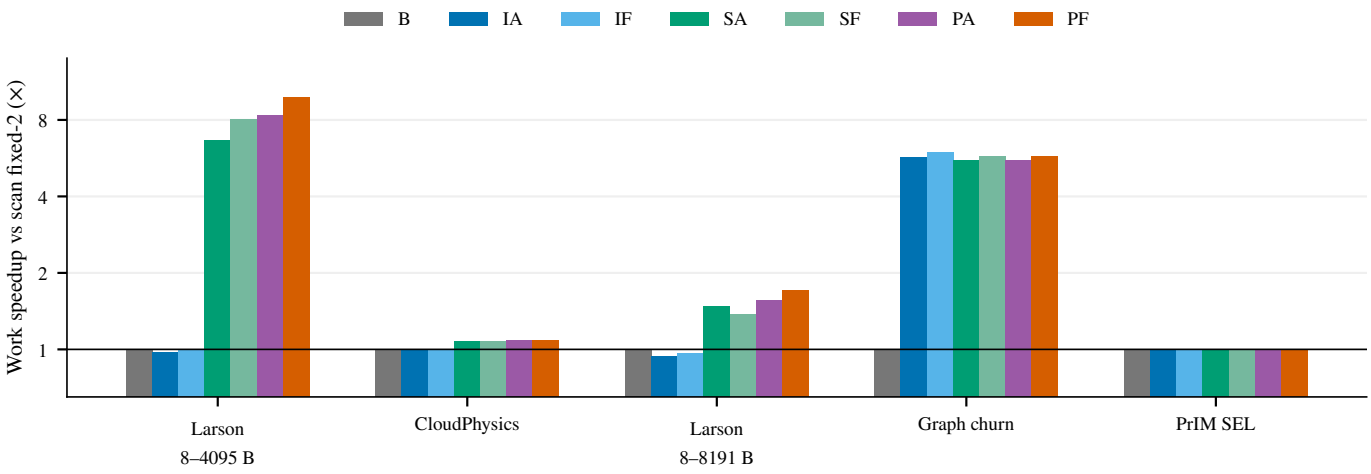

Fig. 1. Every requested final stage, at four tasklets.

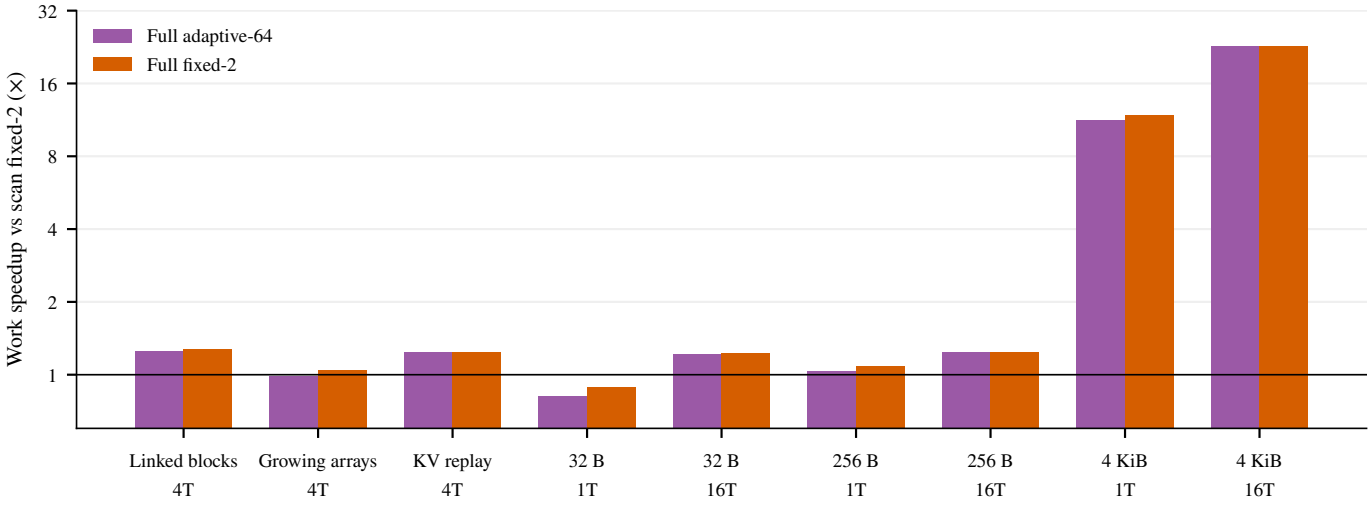

Fig. 2. All final paper-derived/replay settings, including regressions.

TABLE III
PINNING ISOLATED FROM THE SLAB AND RETENTION POLICY.

Workload (4T)	Policy	Work speedup	Total speedup	DRAM-byte reduction (%)
Larson 8–4095 B	A	1.253	1.258	48.34
Larson 8–4095 B	F	1.224	1.230	46.48
CloudPhysics cache	A	1.014	1.011	4.45
CloudPhysics cache	F	1.013	1.010	4.44
Larson 8–8191 B	A	1.056	1.193	66.92
Larson 8–8191 B	F	1.236	1.208	66.49
Gowalla churn	A	0.999	1.169	9.66
Gowalla churn	F	1.000	1.164	9.66
PrIM selection	A	1.000	1.000	0.00
PrIM selection	F	1.000	1.000	0.00

The pinned variant uses a measured 540 B additional WRAM endpoint in each pair. It is an added-storage comparison, not an equal-WRAM-budget replacement experiment.

TABLE IV
ALLOCATION DEMAND, FULL-EXECUTION BACKEND/DRAM ACTIVITY AND PEAK PAYLOAD RESERVATION.

Case	Timed allocs	B buddy	PF buddy	B DRAM MiB	PF DRAM MiB	B peak KiB	PF peak KiB
Larson 8–4095 B (4T)	1,024	1,416	255	5.412	0.722	996	1244
CloudPhysics cache (4T)	150	183	95	5.879	5.365	1408	1596
Larson 8–8191 B (4T)	1,024	1,865	1,311	10.689	2.247	1624	1932
Gowalla churn (4T)	3,397	360	360	15.696	8.433	1440	1440
PrIM selection (4T)	0	0	0	1.876	1.939	0	0
Gowalla linked blocks (4T)	1,687	360	360	7.289	4.409	1440	1440
Gowalla growing arrays (4T)	1,690	93	93	1.858	1.221	372	372
KV growth replay (4T)	1,024	220	220	2.840	1.915	880	880
32 B microbenchmark (1T)	128	2	2	0.032	0.088	8	8
32 B microbenchmark (16T)	2,048	32	32	0.456	0.414	128	128
256 B microbenchmark (1T)	128	9	9	0.114	0.148	36	36
256 B microbenchmark (16T)	2,048	144	144	1.771	1.395	576	576
4 KiB microbenchmark (1T)	128	128	9	1.430	1.086	512	576
4 KiB microbenchmark (16T)	2,048	2,048	144	22.816	16.381	8192	9216

Buddy counts and DRAM bytes cover the full execution; they are not timed-work counts. Peak excludes the extra 64 KiB owner map, buddy metadata and identical application tables. DRAM counters exclude WRAM and instruction fetches.

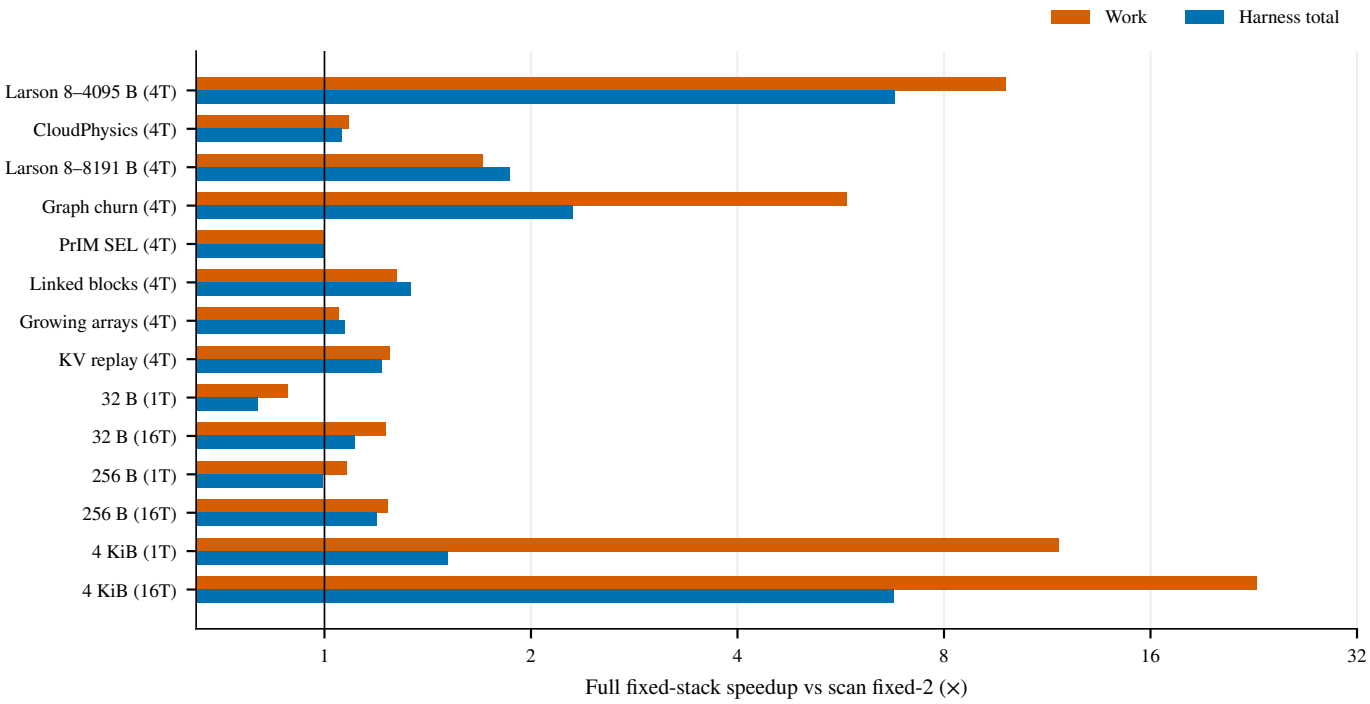

Fig. 3. Work and checked-lifecycle gains differ.